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BRIEF REVIEW

Sleep Architecture Immediately After a Sport-Related Concussion Sustained During a Professional Rugby Union Match: A Retrospective Case Study

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Purpose: A sport-related concussion (SRC) can cause sleep disturbances within 72 hours of injury. However, most research focuses on sleep after >72 hours, with limited attention to acute sleep responses, despite their potential impact on SRC recovery. ***Methods:*** This case study examines a 22-year-old professional rugby union player who sustained an SRC during an away match. Sleep architecture (sleep stages: N1, N2, N3, and rapid eye movement [REM] sleep) and physiological responses during sleep over 4 nights were analyzed—sleep before injury (PRE), the 2 nights immediately following injury (INJ and INJ + 1), and after recovery from injury (POST)—using home-based polysomnography. Return transmeridian travel was undertaken on INJ + 1. ***Results:*** The player took 15 minutes to fall asleep and slept <6 hours on INJ, with reductions in N3 (19% vs 22%–37%) and REM sleep (18% vs 20%–32%) compared with all other nights. Despite increased N3 and REM sleep on INJ + 1, sleep disturbances persisted (23 min to fall asleep, 69 min of wake after sleep onset, 40 awakenings, and sleep efficiency below 85%). Compared with sleep on PRE and POST, mean oxygen saturation and heart-rate variability were lower, mean heart rate was elevated, and respiratory events during sleep were greater on INJ and INJ + 1. ***Conclusion:*** Acute responses after the SRC included sleep disturbances and elevated sympathetic activity (during sleep). The player had difficulty achieving deep, restorative sleep the first night following the SRC, aligning with the non-restorative-sleep complaints commonly reported by concussed athletes on the night(s) immediately following injury.

Keywords: travel, mild traumatic brain injury, collision sports, athlete, sleep

A sport-related concussion (SRC) is a traumatic brain injury caused by an impact to the head, neck, or body leading to neuronal strain.[1] When incurred, SRCs can result in a myriad of acute symptoms such as headaches, sleep disturbances, and cognitive and psychological impairments within 72 hours of injury.[1] Furthermore, up to 70% of concussed individuals experience sleep disturbances,[2] with greater sleep fragmentation linked to prolonged SRC recovery.[3] Anecdotal reports suggest acute phase (≤72 h) symptoms, such as difficulty falling asleep, poor sleep quality, and increased daytime sleepiness,[4,5] but most studies focus on subacute (3–14 d) or chronic (14+ d) responses, leaving the immediate sleep effects poorly understood.[5]

Among the scant existent literature, objective sleep data in the acute phase of injury (SRC) indicate that athletes take longer to fall asleep (16 vs 6 min), exhibit greater variability in sleep patterns across nights, but tend to sleep longer durations compared with controls,[6,7] which partially confirms the subjective reports.[4,5] Notably, the mechanisms for these sleep changes remain unclear. Laboratory-based polysomnography reveals that sleep architecture can change in later phases (≥72 h)[8]; however, traditional lab-based polysomnography is impractical to administer to professional athletes immediately after injury, limiting our understanding of acute SRC sleep mechanisms.[5]

In rugby union, SRCs are the most common match injury, with an incidence of ~19 per 1000 match hours.[9,10] This case study examines the immediate sleep responses (first 2 nights) following an SRC sustained by a player during a match, focusing on objective sleep architecture and physiological responses across a sleep period.

Methods

Participant

A 22-year-old professional male rugby union player (height: 197 cm; weight: 112 kg; position: forward, second-row lock) competing in the Super Rugby Pacific competition was assessed. The player had not sustained any head-related injuries in the preceding 12 months. The player was classified as an intermediate chronotype, with neither a morning nor evening preference based on a validated questionnaire[11] completed ~4 months prior to injury (Supplementary Material [available online]). Informed consent and ethics approval were obtained (2022/ET000700).

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Injury

The player, as the ball carrier, sustained an SRC in a match after a head-to-knee collision with an opposition player. Following video review, medical staff removed the player for an in-game, off-field Head Injury Assessment. The player did not meet the return-to-play criteria and was substituted. Three hours later, a second Head Injury Assessment confirmed the medical diagnosis of an SRC and the player commenced a 14-day return-to-play protocol per World Rugby guidelines. While symptomology cannot be disclosed, the player presented no symptoms of concern and returned to play within the expected timeframe.

Study Design

The study was a retrospective case study.

Methodology

Sleep data from 4 nights were analyzed: PRE (7 nights preinjury), INJ (night of injury), INJ + 1 (second-night postinjury; westward travel across 3 time zones), and POST (24 nights postinjury). A validated home-based polysomnography device (Somfit, Compumedics)[12] recorded sleep, which was uploaded to a proprietary platform (Nexus360; Compumedics) and automatically analyzed.[13] Then, a single sleep scientist manually scored each 30-second epoch into sleep stages (N1, N2, N3, and rapid eye movement [REM] sleep or wake)[14] and corrected artifacts for heart rate (HR), oxygen saturation (SpO_2), and desaturation events. Medication use on these 4 nights was not recorded.

Parameters

- *Sleep:* sleep duration (minutes), sleep efficiency (percentage), onset/offset (hours:minutes), sleep onset latency (minutes), REM latency (minutes), wake after sleep onset (WASO, minutes), awakenings (n), and sleep stages (N1, N2, N3, and REM sleep; percentage and minutes).
- *Physiological:* SpO_2 (percentage), desaturation events (n), HR (beats per minute), heart rate variability (HRV, milliseconds), respiratory events (n), and apnea-hypopnea index (events per hour).

Schedule Details

On PRE (2 d before a home match), the player completed a team training session consisting of a 45-minute gym session and a 75-minute field session. On INJ (match day), the player, named in the starting line-up, played the afternoon away match (kick-off: 15:00 local time), during which he sustained the injury. The following morning, he traveled west across 3 time zones, departing at 09:35 (GMT + 11) and arriving home at 11:15 (GMT + 8). POST was assessed 24 days postinjury, following full recovery, on the night of a training-free day.

Data Analysis

Systematic visual analysis and descriptive statistics across nights are reported.[15] Clinically meaningful changes are also reported to enhance sleep interpretation, using thresholds for sleep quantity (±20 min), sleep onset latency (±10 min), WASO (±20 min), and awakenings (n = ±2).[16]

Results

Sleep Responses

Sleep onset times were 23:47, 23:58, 21:34, and 22:23, and offset times were 04:31, 06:04, 07:14, and 07:15 for PRE, INJ, INJ + 1, and POST, respectively. Clinically meaningful sleep onset latency differences were observed between INJ and INJ + 1 compared with POST, but not PRE (Figure 1d). The number of awakenings during sleep was higher by 11, 31, and 32 on INJ, INJ + 1, and POST, respectively, compared with PRE (all clinically meaningful; Figure 1c). Compared with PRE, WASO was higher by 63 minutes on INJ + 1 and 52 minutes on POST (both clinically meaningful; Figure 1b).

Sleep duration was 261, 332, 489, and 465 minutes on PRE, INJ, INJ + 1, and POST, compared with PRE (all clinically meaningful differences; Figure 2). Sleep efficiency was highest on PRE, with a clinically meaningful difference between PRE and INJ + 1 (Figure 1a). Sleep architecture is displayed in Figure 2.

Physiological Responses

Average SpO_2 during sleep was lowest, and more desaturation events were reported on INJ and INJ + 1, than on PRE and POST (Figure 3c–3e). Mean and maximum HR were higher on INJ and INJ + 1 than on PRE and POST (Figure 3f). Lower HRV was observed on INJ (65.1 milliseconds) and INJ + 1 (66.9 milliseconds) compared with PRE (74.2 milliseconds) and POST (74.0 milliseconds; Figure 3f). The number of respiratory events observed during sleep was higher on INJ, INJ + 1, and POST (69, 104, and 46 events, respectively) compared with PRE (14 events; Figure 3a). The apnea-hypopnea index recorded 3.2, 12.5, 12.8, and 5.9 on PRE, INJ, INJ + 1, and POST, respectively (Figure 3b).

Discussion

Sleep architecture changed on INJ, with the SRC inhibiting the player's ability to achieve deep, restorative sleep (N3), alongside REM sleep, compared with INJ + 1 and POST (Figure 2). This may reflect sympathetic dominance over the parasympathetic nervous system, likely a result of the neurometabolic cascade that SRCs trigger.[17] This heightens hypothalamic-pituitary-adrenal axis activity,[5] which can reduce N3 and REM sleep, and shorten REM latency.[18] Given the importance of N3 and REM sleep for neurological recovery,[1] their reduction may explain the common anecdotal reports of poor sleep quality, impaired cognitive function, and prolonged recovery reported post-SRC.[3]

Elevated mean and maximum HR, reduced HRV, and greater respiratory abnormalities on INJ and INJ + 1 (Figure 3a and 3f) further support sympathetic dominance.[19,20] As seen in sleep apnea patients, elevated sympathetic activity impairs upper airway muscle tone and increases airway collapsibility.[20] Although the assessed player was not diagnosed with sleep-disordered breathing, apnea-hypopnea index scores indicated mild sleep apnea (5–15 events/h) on all nights except PRE, with elevated scores on INJ and INJ + 1 (Figure 3b). Consequently, desaturation events increased (Figure 3e) and mean SpO_2 was lower on INJ and INJ + 1 (Figure 3c and 3d), likely contributing to sleep fragmentation and reduced N3 sleep on INJ. If SRCs exacerbate respiratory abnormalities, this study highlights the need for routine sleep-disorder screenings particularly in collision sport athletes, where sleep-disordered breathing is common.[21]

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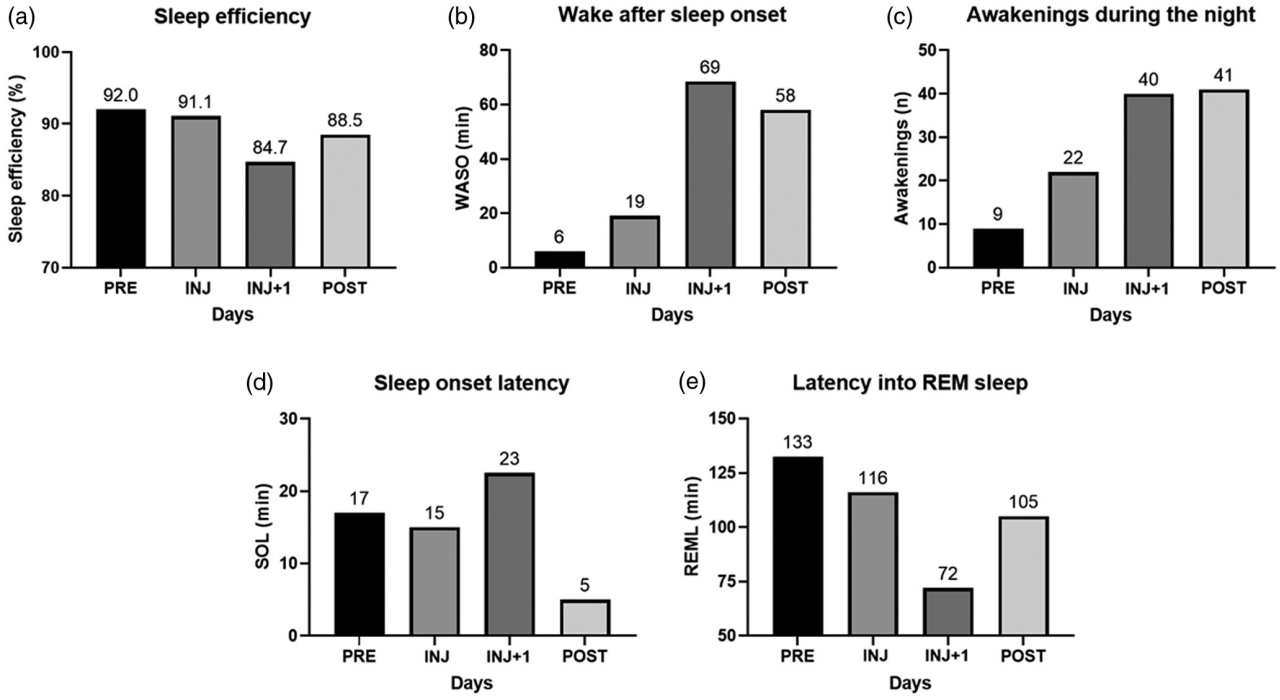

Figure 1 — Sleep characteristics including sleep efficiency (a), wake after sleep onset (b), number of awakenings during the night (c), sleep-onset latency (d), and onset latency into REM sleep (e) before (PRE) and after a sport-related concussion (INJ, INJ + 1, and POST) in a professional male rugby union player (N = 1). Actual values are positioned at the top of each bar. INJ indicates night of injury; INJ + 1, second night postinjury; POST, 24 nights postinjury; PRE, 7 nights preinjury; REM, rapid eye movement; REML, REM sleep latency; SOL, sleep-onset latency; WASO, wake after sleep onset.

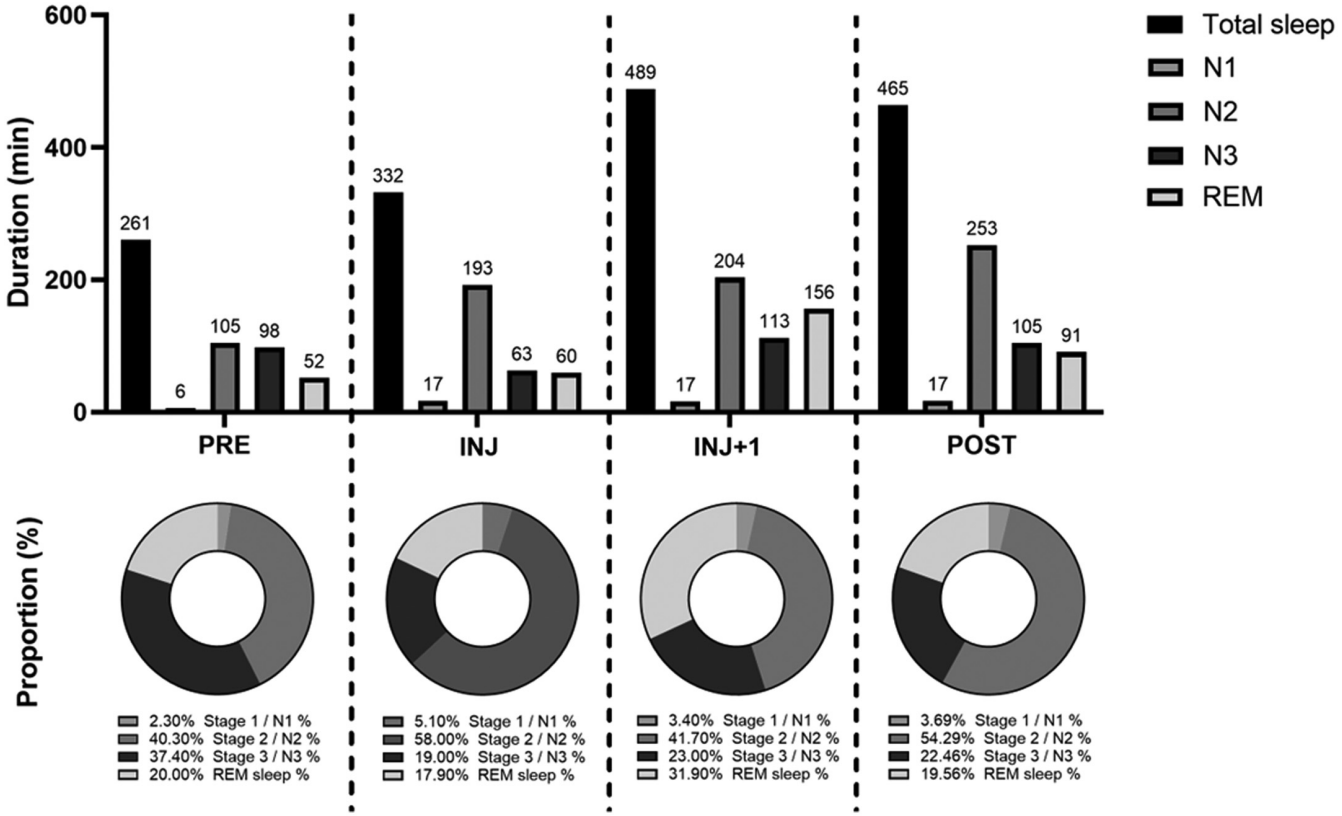

Figure 2 — Sleep architecture before (PRE) and after a sport-related concussion (INJ, INJ + 1, and POST) in a professional male rugby union player (N = 1). The bar graphs on the top illustrate the total time spent asleep and the time spent in each sleep stage (N1, N2, N3, and REM) with values displayed on top of each bar. The pie charts at the bottom show the distribution of sleep stages during a sleep period, represented as a percentage of total sleep duration, with descriptives displayed below each chart. INJ indicates night of injury; INJ + 1, second night postinjury; POST, 24 nights postinjury; PRE, 7 nights preinjury; REM, rapid eye movement.

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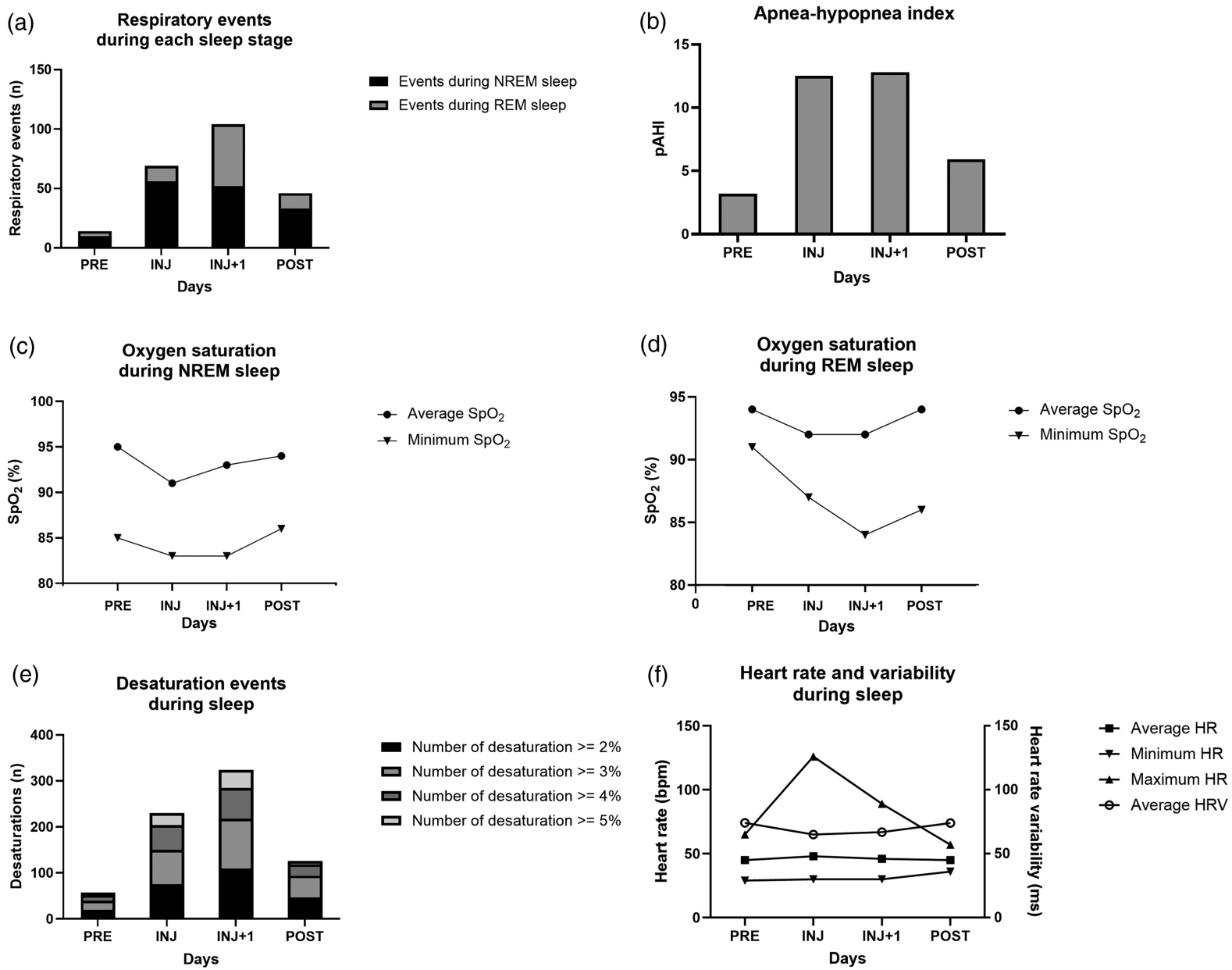

Figure 3 — Physiological measures during sleep, including respiratory measures (a, b), oxygen saturation (c, d), desaturation events (e), and heart rate (f) on nights before (PRE) and after a sport-related concussion (INJ, INJ + 1, and POST) in a professional male rugby union player (N = 1). HR indicates heart rate; HRV, HR variability; INJ, night of injury; INJ + 1, second night postinjury; NREM, non-REM; pAHI, apnea-hypopnea index; POST, 24 nights postinjury; PRE, 7 nights preinjury; REM, rapid eye movement; SpO_2, oxygen saturation.

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On INJ + 1, sleep stages diminished on INJ appear to rebound after westward return travel (Figure 2), with earlier REM sleep onset (Figure 1e). Given that similar travel schedules can alter sleep architecture,[22] transmeridian travel (across 3 time zones) may have compounded sleep disturbances on INJ + 1. Although the shortened REM latency observed on INJ + 1 may reflect heightened hypothalamic-pituitary-adrenal axis activity,[18] the accompanying REM rebound from INJ to INJ + 1 may simply indicate a need to restore REM sleep following its deprivation on INJ. The latter explanation may be more probable, considering N3 sleep nearly doubled from INJ to INJ + 1 as well (Figure 2). While this may indicate deeper, restorative sleep, sleep on INJ + 1 remains fragmented with impaired physiological responses compared with PRE and POST, suggesting ongoing sleep disruption on the second night after injury. Future research should examine INJ + 1 activities, including transmeridian travel, that may further disrupt sleep and SRC recovery.

Finally, while sleep duration increased on INJ and INJ + 1 compared with PRE—similar to existing literature[7]—this likely reflects sleep restriction on PRE (4.4 h) due to an earlier wake time (~04:30) for unknown reasons, rather than an SRC outcome. Theories linking SRC to immediate increased sleep duration often attribute it to greater N3 sleep facilitating glymphatic drainage.[23] However, since N3 sleep decreased on INJ, the observed sleep duration likely compensates for reduced sleep on PRE rather than a true SRC-induced increase. Although it may appear that sleep quality (sleep efficiency, WASO, and awakenings) may be acutely disrupted on INJ and INJ + 1 compared with PRE, as confirmed by existing literature,[6,7] we acknowledge that these alterations could be a result of the shorter sleep duration on PRE. We acknowledge this as a limitation of an opportunistic retrospective case study and recommend that future studies include multiple baseline nights to accurately verify the acute effects of SRC on objective sleep quantity.

Importantly, we acknowledge additional limitations. First, it is unknown whether caffeine, alcohol, and medication were consumed, all of which could influence sleep responses. Future research may consider requesting that the attending physician include a subjective recall of such intake during Head Injury Assessments. Second, jetlag and travel fatigue may have compounded sleep disturbances during the acute injury phase. As travel is common in professional rugby union, future studies should explore whether travel exacerbates SRC symptom severity or prolongs recovery.

Conclusions

This case study supports anecdotal complaints of poor sleep quality during the acute phase (≤72 h) after sport-related concussion (SRC). Immediate alterations in sleep architecture were observed on the night of injury, with reductions in deep, restorative sleep, and REM sleep. Lower oxygen saturation, elevated heart rate, increased respiratory events, and decreased heart-rate variability indicate sympathetic dominance, likely contributing to changes in sleep architecture. These disruptions—potentially exacerbated by transmeridian travel—may contribute to poor sleep quality in concussed athletes, prolonging SRC recovery.

Practical Implications

- SRCs can disrupt objective sleep quality, particularly the maintenance of sleep during the acute phases of injury (≤72 h).
- SRCs may limit deep restorative sleep, as well as REM sleep, potentially contributing to concussion-related symptoms upon waking and delayed return-to-sport timelines.
- Sleep-disorder screenings could be routinely conducted in contact- and collision-sport athletes, as SRCs may exacerbate respiratory sleep disruptions.
- Despite improvements in sleep quantity and architecture by the second night after injury, sleep fragmentation and physiological impairments persist, raising concern for the requirement of transmeridian travel on sleep and SRC recovery if the injury is sustained away from home.

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